Anvil-E1 Phase 2: Official Checkpoint Slice Integration and Software-to-RTL Evaluation

Technical Design and Verification Report

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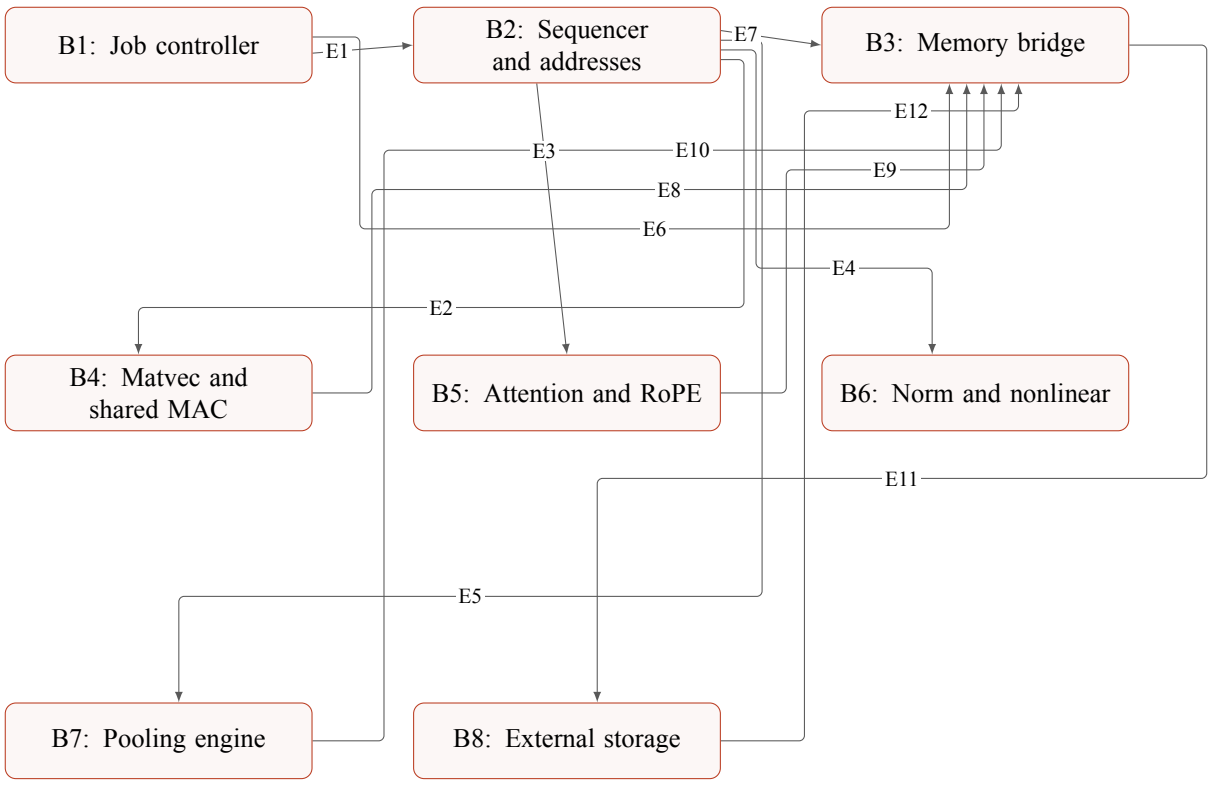
Abstract

Anvil-E1 is a parameterized, text-only SystemVerilog embedding engine with external weight and intermediate storage. Phase 2 integrates authentic BF16 coefficients from the supplied Google EmbeddingGemma-2 checkpoint slice, executes the adopted transformer equations in hardware, and compares public RTL outputs with independent integer and floating-point software references.

All sixteen official-weight jobs passed with zero Q8.24 word mismatches. Against raw-BF16-weight FP32 software inference, minimum cosine similarity was 0.9999999998517652 and maximum absolute error was 0.000012040138244628906. These measurements cover a deterministic reduced geometry with repeated source-layer mapping, not complete-checkpoint inference or application-level embedding quality.

1 Architecture and design rationale

The integration root, anvil_e_top, contains exactly one parameter- and port-forwarding anvil_e1_top instance named core and adds no state or latency. Twelve implementation modules separate control, address generation, memory transport and arithmetic. A hardwired sequencer schedules reusable vector engines and one shared MAC. Weights and sequence-sized intermediates remain external; bounded vector buffers and repeated attention reads avoid an on-chip checkpoint or sequence-squared score array. This favors storage economy and explicit arithmetic boundaries over parallel throughput. Runtime division uses registered restoring arithmetic rather than wide combinational division. Sources E001–E016 describe the implementation; current measurements are supported separately by E085–E089.



Functional block map, not a gate-level netlist. Edge identifiers match the directed relationships below. Crossings do not denote junctions; separate attachment points preserve arrow directions.

Block	Responsibility
B1: Job controller	Latches descriptors, validates input records and address extents, holds embedding words and completion status, and prevents overlapping jobs.
B2: Sequencer and addresses	Schedules the complete forward pass, selects local/full layer geometry, computes packed tensor and scratch addresses, and swaps current/next hidden-state regions.
B3: Memory bridge	Registers requests and responses, permits one outstanding transaction, and preserves payloads through external and internal stalls.
B4: Matvec and shared MAC	Performs row-major matrix-vector operations using signed 64-bit products, a 96-bit accumulator and operation-boundary rounding.
B5: Attention and RoPE	Applies split-half position rotations and grouped-query attention. Repeated external key/value reads implement maximum, denominator and weighted-value passes.
B6: Norm and nonlinear	Provides unit-offset RMSNorm, epsilon-free L2 normalization, GELU, exponential, trigonometric, reciprocal and inverse-square-root operations.
B7: Pooling engine	Computes masked mean embeddings, selects the requested leading channels and normalizes the selected prefix before streaming output.
B8: External storage	Host-managed, nonoverlapping regions hold packed weights, token IDs, absolute positions, masks and sequence-sized scratch.

Directed relationship	Meaning
E1: B1 to B2	Validated job and execution control
E2: B2 to B4	Matrix-vector operation requests
E3: B2 to B5	Attention and rotation requests
E4: B2 to B6	Normalization and nonlinear requests
E5: B2 to B7	Final pooling and output request
E6: B1 to B3	Input-validation reads
E7: B2 to B3	Initialization and intermediate transfers
E8: B4 to B3	Coefficient and vector reads
E9: B5 to B3	Query, key and value transfers
E10: B7 to B3	Projected-token and mask reads
E11: B3 to B8	Registered read/write requests
E12: B8 to B3	Ordered data/error responses

2 Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n	Input	1 bit each	Rising-edge clock and asynchronous active-low reset. Asserted reset forces every public output and payload to zero and cancels active work.
job_valid, job_ready, job_tokens, job_output_dim	Input except job_ready output	1, 1, 32, 16 bits	A descriptor is accepted on job_valid && job_ready at a rising edge. Token count is 1..MAX_SEQ_LEN; output dimension is E, 2E/3, E/3 or E/6. Descriptor fields are latched at acceptance.
weights_base, tokens_base, positions_base, mask_base, scratch_base	Input	64 bits each	Four-byte-aligned byte addresses. Hardware checks exclusive-end overflow; the host guarantees nonoverlap and stable contents during execution.

Signal / group	Direction	Width	Contract
mem_req_valid, mem_req_ready, mem_req_write, mem_req_addr, mem_req_wdata	Output except mem_req_ready input	1, 1, 1, 64, 32 bits	Single-word request channel with one outstanding transaction. Valid and payload remain stable until acceptance.
mem_rsp_valid, mem_rsp_ready, mem_rsp_rdata, mem_rsp_error	Input except mem_rsp_ready output	1, 1, 32, 1 bits	Every accepted request, including a write, requires one ordered response. Response readiness becomes eligible after request acceptance; there is no combinational zero-latency response shortcut.
emb_valid, emb_ready, emb_data, emb_index, emb_last	Output except emb_ready input	1, 1, signed 32, 16, 1 bits	Registered Q8.24 output stream, indexed from zero. Payload and last marker remain stable through stalls. A one-channel result is simultaneously first and last.
done_valid, done_ready, done_error, busy	Output except done_ready input	1, 1, 4, 1 bits	Completion is separately registered and held. busy stays asserted and job_ready stays low through output/completion stalls. Completion acceptance retires the job.

Parameter	Default	Meaning / supported values
VOCAB_SIZE	262144	Positive vocabulary size; token IDs must be smaller.
HIDDEN_SIZE	512	H: 1..65535 hidden channels.
PER_LAYER_INPUT_SIZE	512	P: 1..65535; independent of H.
INTERMEDIATE_SIZE	2048	I: 1..65535 feedforward channels.
EMBEDDING_DIM	768	E: 6..65535, divisible by six; default output dimensions are 768, 512, 256 and 128.
NUM_LAYERS	24	1..65535; full attention at indices congruent to five modulo six.
NUM_Q_HEADS	4	Positive, at most 65535; divisible by both KV-head counts.
LOCAL_KV_HEADS, FULL_KV_HEADS	2, 1	Positive divisors of NUM_Q_HEADS; contiguous query-head groups share a KV head.
LOCAL_HEAD_DIM, FULL_HEAD_DIM	256, 512	Positive even widths, each at most 65535; each NUM_Q_HEADS times head width must also fit 65535.
SLIDING_WINDOW	512	Positive strict absolute-position distance bound.
MAX_SEQ_LEN	262144	Positive sequence-length limit; every supplied position must be below this value.

3 Behavior and corner cases

3.1 Adopted transformer computation

Token embeddings are scaled by sqrt(H). Global PLE projection is divided by sqrt(H), reshaped into distinct per-layer P-vectors and RMS-normalized. Each layer performs input norm; q/k/v projection; per-head q/k norm; RoPE; attention and o projection; post-attention norm and residual; pre-feedforward norm; GELU(gate)*up, down projection, post-feedforward norm and residual; gated PLE projection, post-PLE norm and residual; then layer_scalar. Final RMSNorm and embedding projection precede pooling. Projections are bias-free and inference dropout is zero. These are adopted forward conventions, not demonstrated upstream executable parity.

3.2 Attention, masks and Matryoshka output

Attention is bidirectional. Full layers see all valid keys; local layers additionally require abs(key_position-query_position)<SLIDING_WINDOW. RoPE uses split-half rotation and bases 10000/1000000 for local/full attention. Explicit masks select keys and pooling contributions; no BOS/EOS insertion, automatic PAD

masking or prompt exclusion occurs. Mean pooling precedes prefix selection, and L2 normalization uses only that selected prefix.

3.3 Fixed-point numerical contract

External coefficients, activations and outputs are signed Q8.24 in [-128,128-2^-24]. Finite BF16 conversion and arithmetic boundaries use nearest/ties-even rounding and saturation. Complete MAC reductions retain at least 96 bits; products retain 64 bits. Nonlinear series use Q40 internals. Every weighted RMSNorm adds EPSILON_Q48=281474977 after ties-even division of the complete Q48 square sum by count, then applies unit-offset weights. L2 has no epsilon; a nonempty zero pooled vector succeeds with zero output.

3.4 Arithmetic defect resolution

Quantizing epsilon first to Q8.24 had produced 17/2^24, enough to fail the official FP32 error gate despite exact integer agreement. Direct Q48 epsilon representation corrected that denominator discrepancy. Wide runtime division/remainder and mixed clocked assignment styles were subsequently removed: local 96-bit restoring dividers commit four steps per edge over 24 divide edges, followed by result consumption. The revised design preserves numerical boundaries but increases internal latency. Current functional and structural results cover these repaired sources.

3.5 Errors and reset obligations

Completion errors are 0 success, 1 invalid descriptor, 2 invalid token/position/mask, 3 memory error, 4 all masked and 5 address overflow. IDs are validated even when masked. Exclusive ends above 0xffffffffffffffff are rejected, including an end equal to 2^64. Rejected jobs do not emit a successful embedding. Reset cancels requests, computation and held results; the fabric must share reset or flush obsolete responses.

4 Verification methods and measured results

4.1 Authentic, deterministic official-weight inputs

Method. Streamed SHA-256, parsed raw unique-key BF16 safetensors, validated all 55 tensor descriptors and payload hashes, and independently cross-checked supplied text FP32/Q8.24 coordinates.

Result. Checkpoint digest matched 3a310d7733a28f533b611eb7525f597303980c2ab2f890487a9a1fcb f0366bfb. All 11266 supplied text coordinates passed. Thirty-nine text tensors produced 107 destination tensors and a 5800-byte image; coefficient quantization saturated no values.

Evidence: E085, E035, E002

4.2 Official software-to-RTL numerical agreement

Method. Fresh iverilog -g2012/vvp execution of anvil_e_top; complete public-trace parsing; independent Q8.24 oracle; raw-BF16 FP32 forward inference using struct-rounded scalar arithmetic, sequential reductions and mathematical nonlinearities without hardware clipping.

Result. All 16 jobs and 52 output words passed with zero exact mismatches and no integer-operation saturation. Minimum cosine was 0.9999999998517652 versus >=0.999; maximum absolute error was 1.2040138244628906e-5 versus <=0.002. Worst per-job MAE was 6.884336471557617e-6, RMSE 8.600952306391626e-6 and absolute software/RTL norm difference 7.76098445509632e-8. Every job also passed the 0.002 unit-norm-error gate. The aggregate MAE and RMSE fields are maxima across jobs, not pooled averages.

Evidence: E085, E035, E036, E044

4.3 Official workload and cycle benchmarking

Method. Executed [2], [2,3], [2,4,7] and [2,3,4,5,6,7,8,1], each at dimensions 6/4/2/1. Corresponding positions were [7], [1,2], [1,3,4] and [0..7]; masks were [1], [1,1], [1,0,1] and [1,1,1,1,1,1,0,0].

Result. Descriptor-to-completion cycles, in dimension order 6/4/2/1, were 109606/109484/109364/109304; 246671/246564/246445/246383; 337096/336984/336887/336824; and 1087352/1087221/1087105/1087064. At dimension six, cycles/token were 109606, 123335.5, 112365.3333 and 135919; cycles/valid-token were 109606, 123335.5, 168548 and 181225.3333. Latency is destination accepted rising-edge index minus descriptor accepted edge index, excluding the starting edge and including intervening stalls. The comparison artifact records first/last embedding latencies, tokens/cycle and jobs/cycle separately.

Evidence: E085

4.4 Inherited behavior and independent arithmetic gates

Method. Executed A/B/C at seeds 17, 89 and 203, with exact scoreboards, quantized-input mathematical comparisons, reset cancellation, faults, stalls, invalid descriptors, saturation fixtures and primitive/norm tests.

Result. All nine configuration/seed runs passed. Observed tensor inventories were 107/209/413 for A/B/C, including full layers 5,11,17,23 across the matrix. Nonsaturating inherited accuracy retained its separate 0.02 maximum-error gate. Primitive tests passed GELU <=2e-4, exp <=2e-6, sin/cos <=2e-5 and representable reciprocal/rsqrt relative error <=2e-4. Directed norm tests distinguished Q48 epsilon and preserved zero-L2 handling. The full per-run ledger is retained alongside the official comparison.

Evidence: E085, E031, E033, E038

4.5 Scoreboard sensitivity

Method. A fresh official-weight job first passed the original parser and exact scoreboard. A private trace copy changed one accepted output word by one Q8.24 LSB without changing protocol events.

Result. The identical scoreboard rejected precisely job 0, index 0: expected 11924784, observed 11924785. The original trace passed; the altered trace was not counted as a successful DUT execution. The inherited functional negative control also passed.

Evidence: E086, E085

4.6 Packing, defaults and geometry boundaries

Method. Default elaboration, wrapper/interface checks, deterministic address comparisons, malformed-header rejection and exhaustive BF16 conversion checks.

Result. Validated 413 default tensors, 65280 finite BF16 encodings, rejection of 256 nonfinite encodings and seven illegal-parameter cases. Address checks covered default/A/B/C. Default weight extent was 1084010592 bytes; maximum configured default scratch was 18253611008 bytes.

Evidence: E088, E003, E016

4.7 Synthesizable connected datapath without runtime division hazards

Method. Fresh Yosys synthesis of A and B through hierarchy, process, optimization, memory and consistency passes, followed by source-bound timing-style/netlist inspection.

Result. Both configurations passed hierarchy/connectivity, no-latch and no-unresolved-DUT assertions. A contained 272 sequential and 204 arithmetic cells; B contained 355 and 215. Both retained all thirteen modules and contained zero combinational division/remainder cells. Timing-style inspection passed; these are generic structural results, not physical timing closure.

Evidence: E087, E089, E042, E043

5 Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

5.1 Official Weights Software-Layer vs Hardware-RTL / FPGA Comparison

Metric	Measured Value	Notes
Evaluated Official-Weight Jobs	16	0 bit mismatches vs integer oracle
Minimum Cosine Similarity (FP32 vs Hardware)	0.999999999852	Threshold >= 0.999
Maximum Absolute Error (FP32 vs Hardware)	1.204014e-05	Fixed-point Q8.24 quantization bound
Mean Absolute Error / RMSE	6.884336e-06 / 8.600952e-06	Across all output dimensions
Exact Word Mismatches (RTL vs Reference)	0	100% bit-exact hardware match

5.2 Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Gate	Status	Measured Summary
L1 Static RTL Timing & Synthesis Lint	passed	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	passed	WNS_setup = 7.916 ns, failing setup endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0

6 Integration and use

Use rtl/anvil_e_top.sv with all rtl/*.sv; direct anvil_e1_top compatibility remains available. Read design/ARCHITECTURE.md for packing, scratch layout and register boundaries, and design/OFFICIAL_WEIGHTS_EVALUATION.md for official mapping.

Official configuration A is (V,H,P,I,E,L,Q,localKV,fullKV,localD,fullD,window,maxSeq)=(16,4,4,8,6,6,2,1,1,2,4,2,8). Leading matrix rows/columns retain loaded-source row stride. Destination layers 0..4 repeat official layer 0; destination layer 5 uses official layer 5. Global PLE rows 0..23 supply distinct four-row layer inputs, including rows 20..23 for destination layer 5.

Pack globals lexically, then ascending destination layer and lexical suffix, using row-major little-endian Q8.24 words. The official image SHA-256 is 2cf8db510462a09990f2d66c345984866d0d6374d6997b7fa4c203a9e5e1e147. Mapping and inventory identities accompany evidence/verification/functional/official_weights_comparison.json.

For a complete default checkpoint, use python3 design/pack_model.py --input /supplied/model.safetensors --output weights.q24.bin --inventory weights.inventory.json with existing output parent directories. This strict 413-tensor packer intentionally rejects the supplied partial official slice.

Allocate scratch bytes as 4*N*(2*H+L*P+(Q+2*max(localKV,fullKV))*max(localD,fullD)). Initialize only coefficients, IDs, positions and masks; provide writable scratch. Honor every response, output and completion handshake.

Reproduction dependencies are Python 3 standard library, iverilog/vvp, Yosys and the supplied immutable input files; NumPy is unnecessary. The delivered entrypoint is tests/verify.py. For example: python3 tests/verify.py --check functional --output evidence/verification/functional --timeout-s 82800 --configs A,B,C --seeds 17,89,203. Sensitivity and synthesis are separate checks. No clean-directory portable-reproduction experiment is claimed.

7 Limitations and acceptance boundary

Official evaluation is a reduced authentic slice, not original-geometry 24-layer inference. Additional exercised synthetic geometries were B=(20,6,4,10,12,12,4,2,1,2,4,3,8) and C=(16,4,4,8,6,24,4,2,1,2,4,2,4), in the integration tuple order. Default elaboration does not establish default-scale inference performance.

Excluded supplied tensors are embed_vision.embedding_projection.weight; vision_tower.patch_embedder.input_proj.weight and position_embedding_table; and vision_tower.encoder.layers.0 tensors input_layernorm.weight, post_attention_layernorm.weight, post_feedforward_layernorm.weight, pre_feedforward_layernorm.weight, mlp.down_proj.linear.weight, mlp.gate_proj.linear.weight, mlp.up_proj.linear.weight, self_attn.k_norm.weight, self_attn.q_norm.weight, self_attn.k_proj.linear.weight, self_attn.o_proj.linear.weight, self_attn.q_proj.linear.weight and self_attn.v_proj.linear.weight. Their extents and hashes were validated, but none executes. Audio is excluded.

The software comparison uses BF16 source coefficients decoded exactly into FP32, with FP32 activation arithmetic; it is not a BF16-activation benchmark. Local provenance checks establish consistency with supplied manifests, not independent remote checkpoint retrieval.

Cycle-based throughput includes the exercised memory and backpressure behavior. The 10 ns testbench period is only a simulation convention. No physical frequency, FPGA implementation, board execution, technology area, power, timing signoff, full-model semantic quality or upstream executable parity is established.

A Source inventory

Production inventory: 13 Verilog/SystemVerilog files; 2728 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
rtl/anvil_e1_address.sv	187	176
rtl/anvil_e1_attention.sv	335	332
rtl/anvil_e1_controller.sv	179	175
rtl/anvil_e1_mac.sv	34	34
rtl/anvil_e1_matvec.sv	99	99
rtl/anvil_e1_memory.sv	65	65
rtl/anvil_e1_nonlinear.sv	238	235
rtl/anvil_e1_norm.sv	205	203
rtl/anvil_e1_pool.sv	198	197
rtl/anvil_e1_rope.sv	182	182
rtl/anvil_e1_sequencer.sv	817	801
rtl/anvil_e1_top.sv	158	158

Production file	Physical lines	Code lines
rtl/anvil_e_top.sv	71	71

B Verification record index

Recorded check	Outcome	Execution time (s)
functional	passed	961.712
sensitivity	passed	15.268
synthesis	passed	75.840

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E002	design/OFFICIAL_WEIGHTS_EVALUATION.md
E003	design/pack_model.py
E016	rtl/anvil_e_top.sv
E031	tests/behavior.py
E033	tests/model.py
E035	tests/official.py
E036	tests/official_fp32.py
E038	tests/primitives.py
E042	tests/synthesis.py
E043	tests/timing.py
E044	tests/verify.py
E085	.vulcan/engineering/run-0003/attempt-0018/stdout.log
E086	.vulcan/engineering/run-0003/attempt-0019/stdout.log
E087	.vulcan/engineering/run-0003/attempt-0020/stdout.log
E088	.vulcan/engineering/run-0003/attempt-0021/stdout.log
E089	.vulcan/engineering/run-0003/attempt-0022/stdout.log

C Document provenance

Engineering was independently accepted for the recorded source and evidence identity. This is not exhaustive proof, hardware deployment or physical signoff. The companion REPORT.md records the exact report approval and source manifest, and links the English/Chinese journeys containing original failures and interventions. A report number identifies a generation attempt, not a new hardware revision.

No developer attribution was supplied; none has been inferred.